Hot Axle Boxes & Rotational Tests — Revision Guide

Module 3 — Train Working Module | DC63 Driver Plan, Week 6 (22-26 June 2026)

All rules and speeds in this guide are taken only from GERT8000-TW5. Descriptive/technical content and photos are from the EMR “Hot 100-10: Axle boxes & Rotational tests” training slides.

1. What Is a Hot Axle Box?

An axlebox houses the bearing that allows a wheelset to rotate. If a bearing starts to fail it generates excess friction and heat — a “hot axlebox.” Left undetected, the bearing can fail completely and cause a derailment.

A hot axlebox can be picked up in three ways:

- lineside detection equipment (a Hot Axlebox Detector, HABD)
- a built-in onboard detection system fitted to the vehicle
- a member of staff noticing it (e.g. smoke, smell, or visible damage)

This subsection comes from the classroom slides (PPTX), not from TW5 itself. TW5 states the operational rule only — it does not explain what a hot axlebox is or how it is caused. Included here for background understanding.

Close-up of an axlebox and bearing assembly.

2. How Hot Axle Boxes Are Detected

Hot Axlebox Detectors (HABD) are lineside sensors positioned at intervals along the route. As a train passes, the sensors check the temperature of each axlebox. If a reading is too high, an alarm is given.

This subsection comes from the classroom slides (PPTX), not from TW5 itself. TW5 states the operational rule only — it does not explain how the detection equipment works. Included here for background understanding.

A lineside Hot Axlebox Detector (HABD) unit beside the track.

3. Starting a Journey — Hot Axleboxes & Defective Bearings

You must not start a journey if you are aware that a vehicle has:

- a hot axlebox
- a defective axle bearing, or
- a defective onboard detection system

GERT8000-TW5 §16.1

4. A Hot Axlebox Develops During a Journey

If a vehicle develops a hot axlebox while on a journey, you must:

- stop the train as soon as possible
- tell the signaller, including whether the train is conveying dangerous goods
- not allow the train to move again until told to do so, and then carry out the signaller's instructions
- if you are a driver of a DO train, arrange for passengers to be transferred before the train is moved, if practicable
- get authority from the RST if you are in any doubt about how to proceed
- if the train is moved before the vehicle has been examined, not exceed 10 mph (15 km/h), or 5 mph (10 km/h) over points or crossings

GERT8000-TW5 §16.2

5. Examining the Train After a Lineside HABD Activation

If you are not able to start examining the vehicle within 10 minutes of the train being stopped, you must tell the signaller and follow their instructions. If the train is then moved before the examination has taken place, you must not exceed 20 mph (30 km/h).

GERT8000-TW5 §16.3(c)

When you check for evidence of overheating, you must examine the vehicle identified. If there is no evidence of overheating, you must also check the other axleboxes on both sides of that vehicle, and on the vehicles on either side of it. You must tell the signaller of any defects found.

GERT8000-TW5 §16.4

6. No Evidence of Overheating

If there is no evidence of overheating, and all of the vehicles you examined have roller bearings, the train can proceed normally.

However, you must not move the train until instructed by the signaller, and then follow their instructions, if either:

- the train has been stopped because of another HABD activation within the previous 50 miles (80 km), or
- any of the vehicles examined has a bearing other than a roller bearing

If the train is moved in these circumstances, you must not exceed 20 mph (30 km/h).

GERT8000-TW5 §16.5

The same principle applies after an onboard detection system activation: if there is no evidence of overheating, the train can proceed normally — unless there is a second activation of the same onboard system, or a lineside HABD activation occurs. In that case you must not move until instructed by the signaller, follow their instructions, and if moved, not exceed 20 mph (30 km/h).

GERT8000-TW5 §16.8

7. Evidence of Overheating

If there is evidence of overheating, you can move the train to the next location where it can be dealt with appropriately. You must:

- get authority from the RST if you are in any doubt about how to proceed
- get the signaller’s authority before the train is moved
- if you are a driver of a DO train, arrange for passengers to be transferred before the train is moved, if practicable
- not exceed 10 mph (15 km/h), or 5 mph (10 km/h) over points or crossings

GERT8000-TW5 §16.6

Evidence of overheating on an axlebox.

8. Onboard Detection System Activation

If an onboard detection system activates, you must:

- tell the signaller immediately
- unless RST are immediately available, examine the axlebox identified by the system
- get the signaller's authority before the train is moved
- get authority from the RST if you are in any doubt about how to proceed
- if you are a driver of a DO train, arrange for passengers to be transferred before the train is moved, if practicable
- not exceed 10 mph (15 km/h), or 5 mph (10 km/h) over points or crossings

GERT8000-TW5 §16.7

9. Checking Temperature: the Tempilstik

A Tempilstik is a heat-sensitive crayon used to check axlebox temperature. The version used for this check is rated at 169°F (76°C).

When the mark is applied to a heated surface, it will melt and a liquid smear will develop if the surface is at or above the rated temperature. If no liquid smear is evident, the axlebox temperature is normal.

This subsection comes from the classroom slides (PPTX), not from TW5 itself. TW5 states the operational rule only — it does not describe the Tempilstik tool. Included here for background understanding.

A Tempilstik temperature-indicator crayon.

Tempilstik mark being applied to an axlebox cap.

10. Starting a Journey — Locked Wheels, Wheel Flats, Shifted Tyres & Dragging Brakes

You must not start a journey if you are aware that a vehicle has:

- locked wheels
- shifted tyres
- dragging brakes, or
- obvious wheel damage as described in §27.2(c) (see Section 12 below)

GERT8000-TW5 §27.1

11. Dragging Brakes & the Wheel Rotational Test

If you find a brake dragging, you must:

- attempt to release the brake locally
- examine the brakes, tyres and wheels for damage or overheating
- if there is evidence of damage, follow the instructions in §27.2(c) (see Section 12 below)
- if the brake cannot be fully released, not exceed 10 mph (15 km/h), or 5 mph (10 km/h) over points or crossings

GERT8000-TW5 §27.2(a)

After freeing locked wheels, you must make sure that the wheels will rotate freely before you proceed — this check is the wheel rotational test.

GERT8000-TW5 §27.2(b)

In practice, a rotational test is carried out either:

- with assistance — a competent crew member or fitter watches the wheel while, with the signaller's permission, the train is moved forward a short distance to confirm the wheel turns freely, or
- without assistance — the suspected wheel is marked with chalk or ballast and compared against an unaffected wheel after a short, low-speed movement. Freight trains also carry out a roll-by test on departure from sidings.

This procedure comes from the classroom slides (PPTX), not from TW5 itself. TW5 sets the rule (wheels must rotate freely before proceeding) but does not describe how the test is carried out. Included here for background understanding. A vehicle with locked wheels must not be moved.

12. Checking for Damage to the Wheels

After checking for locked or hot wheels, you must continue only as shown in the following table:

Wheels freed?	Condition found	Action required
Yes	No evidence of damage	Train can proceed normally; tell the train operator's control at the first convenient opportunity.
Yes	Obvious damage — flat approximately 60-100mm	Report to signaller immediately; do not move until instructed; follow instructions; if moved, not exceed 20 mph (30 km/h) unless instructed otherwise by RST.
Yes	Serious damage: flat longer than 100mm; flat forming a flange; suspected shifted tyre	Report to signaller immediately; do not move until examined by RST; do not move until instructed by signaller; follow instructions.
No	Any condition	Report to signaller immediately; do not move until examined by RST; do not move until instructed by signaller; follow instructions.

GERT8000-TW5 §27.2(c)

A normal, undamaged wheel profile.

A wheel showing flat damage on the tread.

13. Doubt or Obvious Damage

If there is doubt that the train can proceed safely, you must tell the signaller immediately and not move the train until it has been examined by the RST.

> *GERT8000-TW5 §27.2(d)*

If the damage to the vehicle is obvious, you must tell the signaller immediately.

GERT8000-TW5 §27.2(e)

14. Emergency Bypass Switch (EBS)

The EBS allows traction power to be obtained despite an open door or incomplete door-proving circuit. It is used as a last resort.

This introductory line comes from the classroom slides (PPTX), not from TW5 itself. TW5 states the operational rule only — it does not explain what the EBS is for. Included here for background understanding.

You must not start a journey if the EBS has been operated in any cab.

GERT8000-TW5 §12.1

The train can start, not carrying passengers, to travel to a maintenance depot, if you tell the signaller, get their permission, and tell the guard.

GERT8000-TW5 §12.2

If you need to operate the EBS during a journey, you must:

- tell the signaller immediately
- not move until instructed, and then follow their instructions
- tell the guard if the train is moved
- if the guard is not able to travel in the rear of the train, arrange for a competent person to do so if possible

GERT8000-TW5 §12.3

15. Traction Interlock Switch (TIS)

The TIS prevents traction power being obtained if a door is open. Like the EBS, it can be overridden as a last resort.

This introductory line comes from the classroom slides (PPTX), not from TW5 itself. TW5 states the operational rule only — it does not explain what the TIS is for. Included here for background understanding.

You must not start a journey if the TIS has been operated or is unsealed in any cab. You must also not start a journey carrying passengers if the TIS has been operated.

GERT8000-TW5 §24.1, §24.2

You must only operate the TIS when the train is at a stand and unable to get traction power, after checking that all doors are closed on both sides of the train. You must then tell the signaller immediately, not move until instructed, tell the guard, and follow the signaller’s instructions.

GERT8000-TW5 §24.3

When the journey is over, you must restore the TIS to the normal position before shutting down the driving controls if the train is being stabled, reversed, or coupled to another train driven from another cab. You must not leave the switch in the isolate position in any cab other than the one you are driving from, except where the TIS can only be restored by the RST.

GERT8000-TW5 §24.5

16. Brake Continuity Test

If the brake-pipe coupling heads separate and the train stops, you may try to recouple them if undamaged. If recoupled, the train may continue normally as long as you tell the signaller and carry out a brake continuity test.

GERT8000-TW5 §5.2

This trigger note comes from the classroom slides (PPTX), not from TW5 itself: after dealing with any irregular operation of the brake (including a dragging brake), a brake continuity test should be undertaken, and some companies also require a rotational test after a HABD detection. Included here for background understanding.

Revision Questions

Q1. Name the three ways a hot axlebox can be picked up.

Answer: Lineside detection equipment (HABD), a built-in onboard detection system, or a member of staff noticing it.

EMR training slides (classroom material)

Q2. You become aware before starting a journey that a vehicle has a hot axlebox. What must you do?

Answer: You must not start the journey.

GERT8000-TW5 §16.1

Q3. A vehicle develops a hot axlebox during a journey. List four things you must do.

Answer: Stop the train as soon as possible; tell the signaller (including dangerous goods status); do not move again until told, then follow instructions; if a DO train driver, arrange passenger transfer if practicable before moving; get RST authority if in doubt.

GERT8000-TW5 §16.2

Q4. If the train must be moved before a hot axlebox examination has taken place, what speed limit applies?

Answer: 10 mph (15 km/h), or 5 mph (10 km/h) over points or crossings.

GERT8000-TW5 §16.2

Q5. You can't start examining the vehicle within 10 minutes of stopping. What must you do, and what speed applies if the train is then moved before examination?

Answer: Tell the signaller and follow their instructions; if moved before examination, not exceed 20 mph (30 km/h).

GERT8000-TW5 §16.3(c)

Q6. When examining a vehicle identified by a HABD, what else must you check if you find no evidence of overheating on that vehicle?

Answer: The other axleboxes on both sides of that vehicle, and on the vehicles either side of it; tell the signaller of any defects found.

GERT8000-TW5 §16.4

Q7. No evidence of overheating is found and all examined vehicles have roller bearings — can the train proceed normally?

Answer: Yes.

GERT8000-TW5 §16.5

Q8. Give two circumstances where, even with no evidence of overheating, you must not move until instructed by the signaller.

Answer: The train was stopped because of another HABD activation within the previous 50 miles (80 km); or any examined vehicle has a bearing other than a roller bearing.

GERT8000-TW5 §16.5

Q9. There is evidence of overheating. What must you do before the train is moved?

Answer: Get the signaller's authority (and RST authority if in doubt); if a DO train driver, arrange passenger transfer if practicable.

GERT8000-TW5 §16.6

Q10. What is a Tempilstik rated at, and what indicates overheating when it is applied?

Answer: 169°F (76°C). If the surface is at or above this, the mark melts and a liquid smear develops — no smear means a normal temperature.

EMR training slides (classroom material)

Q11. After freeing locked wheels, what must you confirm before proceeding?

Answer: That the wheels will rotate freely — the wheel rotational test.

GERT8000-TW5 §27.2(b)

Q12. A wheel has a flat of approximately 80mm and can be freed. What must you do?

Answer: Report to the signaller immediately; do not move until instructed; follow instructions; if moved, not exceed 20 mph (30 km/h) unless instructed otherwise by RST.

GERT8000-TW5 §27.2(c)

Q13. The wheels cannot be freed at all. What must you do, regardless of condition?

Answer: Report to the signaller immediately; do not move until examined by RST and instructed by the signaller; follow instructions.

GERT8000-TW5 §27.2(c)

Q14. What is the EBS used for, and what must you do before starting a journey regarding it?

Answer: It allows traction power despite an open door or incomplete proving circuit, used as a last resort. You must not start a journey if it has been operated in any cab.

GERT8000-TW5 §12.1

Q15. When can you operate the TIS, and what must you check first?

Answer: Only when the train is at a stand and unable to get traction power, after checking all doors are closed on both sides of the train.

GERT8000-TW5 §24.3

Q16. The brake-pipe coupling heads separate and the train stops. You recouple them undamaged. What must you do before continuing normally?

Answer: Tell the signaller and carry out a brake continuity test.

GERT8000-TW5 §5.2